

WATERLOO REGION · DATA FOR GOOD · CHRONIC HOMELESSNESS ML

Phase B Report

Service Features & Intervention Outcomes

Model 1 retrained with four new service features derived from CAM data — housing type and intake intervention — achieving a significant lift in predictive performance. A new descriptive analysis answers PECH's original question about which interventions are associated with better outcomes in high-risk clients.

Date July 2026 **Dataset** D4G_CAM_Final.xlsx **Valid clients** 9,576 **Model** LightGBM · Optuna 50 trials

Train / Test split Feb 1, 2024

01 Context and Objectives

Phase A delivered a working risk model (19 features, ROC-AUC 0.779) that flags clients likely to become chronically homeless at the moment of intake. Phase B had two objectives:

1. **Extend Model 1** with service features already present in the CAM data — specifically the client's housing type and first intervention at intake — to improve predictive accuracy.
2. **Answer PECH's original question:** which interventions are associated with avoiding chronic homelessness, particularly among clients the model identifies as high-risk?

DESIGN CONSTRAINT

All new features are derived from the client's *first* snapshot only, preserving the no-leakage guarantee of the Phase A model. The model still predicts risk at the moment of intake using only information available on day one.

02 Four New Features

Phase B adds four features to Model 1, bringing the total from 19 (Phase A) to 23. All four are extracted from the first CAM snapshot per client.

FEATURE	SOURCE COLUMN	TYPE	NOTES
last_known_housing_category NEW	Last Known Housing Type	Categorical · 9 values	63 raw values grouped into 9 stable categories
last_known_housing_missing NEW	Last Known Housing Type	Binary	1 if NaN or "None Specified" (22.3% of records)
first_intervention_type NEW	Recent Interaction Module	Categorical · 10 values	Single categorical replaces 8 mutually exclusive binary flags
days_since_last_activity NEW	Days Since Last Activity	Numeric	2.8% missing · measures engagement gap at intake

Housing type grouping (63 → 9 categories)

The raw field contains 63 distinct values created by caseworkers over time (duplicate names, abbreviations, provider suffixes). We consolidated them into 9 meaningful categories based on

housing stability and service type. The grouping was reviewed and approved by the technical lead.

CATEGORY	CLIENTS	%
Emergency Shelter	16,170 rows	27.1%
Unknown (NaN + None Specified)	13,294	22.3%
Street / Encampment	6,643	11.1%
Cambridge Shelter STANDALONE	6,221	10.4%
Motel	5,864	9.8%
Couch Surfing / Family	4,000	6.7%
Stable Housing	3,550	6.0%
Transitional Housing	2,784	4.7%
Institutional	1,098	1.8%

Cambridge Shelter is kept as a standalone category because it exceeds 10% volume — the threshold recommended by the technical lead to avoid grouping a dominant provider into a generic bucket.

First intervention type distribution

INTERVENTION AT INTAKE	CLIENTS	SHARE	CHRONIC RATE
Goods and Services	4,439	46.2%	17.5%
Admissions	3,105	32.4%	56.5%

Group Activities	773	8.1%	33.0%
Case Management	676	7.1%	47.8%
Client Details	431	4.5%	16.5%
SPDAT	62	0.6%	71.0%
Service Restrictions	46	0.5%	56.5%
Housing Placement	22	0.2%	27.3%
Other / missing	22	0.2%	—

SELECTION BIAS — IMPORTANT

Chronic rates differ widely across interventions, but this reflects who gets assigned each intervention, not whether the intervention works. SPDAT (71%) and Case Management (48%) are assigned to the most vulnerable clients from the start. The model accounts for this through the mean risk score — not through raw chronic rates.

03 Model Performance — Phase A vs Phase B

Both models use an identical temporal split: clients whose first record was created on or before February 1, 2024 form the training set; those after form the test set. The Phase A baseline was retrained on the same split for a fair comparison.

TEST SET SIZE	CHRONIC RATE — TEST	PHASE B ROC-AUC	RECALL @0.20
2,349	17.9%	0.874	0.983

clients (after Feb 2024)	vs 39.3% in training set	▲ +0.129 vs Phase A	chronic class · Phase B
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ROC-AUC comparison

Phase A	0.744
19 features	
Phase B	0.874
23 features	

Full metrics at threshold 0.20 (production setting)

METRIC	PHASE A (19 FEAT.)	PHASE B (23 FEAT.)	CHANGE	TARGET
ROC-AUC	0.744	0.874	+0.129	> 0.82 ✓
Recall (chronic class)	0.542	0.983	+0.442	> 0.75 ✓
Precision (chronic class)	0.399	0.377	-0.022	> 0.50 ✕
F1 score	0.459	0.545	+0.086	> 0.65

What the metrics mean in plain terms

The test set contains **2,349 clients**, of whom **421 actually became chronic** (17.9%). Here is what each metric means in concrete terms at threshold 0.20:

METRIC	WHAT IT MEASURES	IN PRACTICE (PHASE B)
ROC-AUC 0.874	If you pick one chronic client and one non-chronic client at random, how often does the model rank the chronic one higher?	87.4% of the time the model correctly ranks the higher-risk client above the lower-risk one.

Recall 0.983	Of all clients who <i>actually</i> became chronic, how many did the model flag?	Of the 421 truly chronic clients, 414 are correctly flagged . Only 7 slip through undetected.
Precision 0.377	Of all clients the model flags as high-risk, how many actually became chronic?	The model flags ~1,098 clients as high-risk. Of those, 414 truly become chronic and ~684 are false alarms — caseworkers review them and no action is needed.
F1 0.545	Balanced average of recall and precision.	Improves from 0.459 (Phase A) to 0.545 — both recall and precision moved in the right direction together.

THE RECALL-PRECISION TRADE-OFF

Setting the threshold low (0.20) casts a wide net: the model catches **98.3% of truly chronic clients**, at the cost of **~6 in 10 alerts being false alarms**. Raising the threshold would reduce false alarms but miss more real cases.

For this use case — social services where **missing a high-risk client is more costly than a false alarm** — high recall is the right priority. A caseworker can review an alert that turns out to be unnecessary; they cannot help a client who was never flagged.

The housing features are the dominant driver of the Phase B lift.

`last_known_housing_category` is the single most predictive feature in the model — consistent with Phase A's finding that housing stability is the strongest predictor of chronic homelessness. The intervention feature (`first_intervention_type`) contributes meaningfully but ranks lower, in part because it captures assignment patterns (who gets which intervention) rather than intervention effects.

04 Intervention Outcomes Analysis

Notebook 06_intervention_outcomes.ipynb addresses PECH's original question descriptively: among clients the model identified as high risk at intake, which interventions were present in those who ultimately avoided chronic homelessness?

OBSERVATIONAL DATA — ASSOCIATIONS ONLY

This analysis cannot establish causality. Differences between groups may reflect who was assigned each intervention (initial risk profile), not the direct effect of the intervention itself. The findings are a guide for generating hypotheses — not proof of effectiveness.

The high-risk group

3,850 clients (40.2% of all valid clients) received a Model 1 score above 0.50 — classified as high risk at intake. Within this group:

HIGH RISK TOTAL	NEVER BECAME CHRONIC ✓	DID BECOME CHRONIC ✗	MEDIAN DAYS IN SYSTEM
3,850	1,147	2,703	90
score > 0.50 at intake	29.8% of high-risk	70.2% of high-risk	all valid clients

Intake intervention by group

How interventions distribute within each subgroup (each column sums to 100%). The difference column shows the gap between the two groups in percentage points.

NEVER BECAME CHRONIC ✓ (N=1,147)		DID BECOME CHRONIC ✗ (N=2,703)	
Goods and Services	28.9%	Goods and Services	19.9%
Admissions	26.1%	Admissions	27.4%

Case Management	19.0%	Case Management	25.7%
Group Activities	12.5%	Group Activities	13.8%
Client Details	6.5%	Client Details	5.6%
Other	7.0%	Other	7.6%

Key patterns

GOODS AND SERVICES SIGNAL

+9 percentage points at intake (28.9% vs 19.9%) and **+11 points at last recorded intervention** (31.3% vs 20.2%) in clients who never became chronic. This is the most consistent signal across both intake and trajectory analyses. Mean risk score for Goods and Services clients is 0.712 — substantially lower than Case Management (0.869), but still in the high-risk band.

CASE MANAGEMENT AND SELECTION BIAS

Case Management appears much more frequently in clients who did become chronic (25.7% vs 19.0% at intake; 16.6% vs 6.1% at last intervention). This is consistent with **selection bias**: caseworkers assign intensive case management to the most vulnerable clients, who have the highest initial risk regardless of the intervention. Mean risk score for Case Management clients is **0.869** — the highest of all interventions.

Trajectory analysis

Looking beyond the intake snapshot, we built the full intervention sequence for each client across all their CAM records:

METRIC	NEVER BECAME CHRONIC 	DID BECOME CHRONIC 
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Avg. snapshots per client	~3.5	~7.1
% who changed intervention	lower	higher
Avg. distinct interventions	lower	higher

Clients who became chronic had more snapshots and more intervention changes — consistent with longer, more complex cases. The "never chronic" group left the system faster and with fewer touchpoints.

Most common intervention transitions

In high-risk clients who never became chronic, the most frequent consecutive intervention pairs are:

- Admissions → Goods and Services (88 instances)
- Admissions → Group Activities (58 instances)
- Goods and Services → Admissions (53 instances)

In clients who did become chronic, the same transitions appear but at much higher volume, with Case Management transitions becoming far more prominent (Admissions → Case Management: 355 instances).

05 Complete Feature Set — 23 Features

entry_year	entry_season	days_before_first_snapshot	age	gender	indigenous_status
veteran_status	immigration_status	household_type	has_income	income_source	yearly_income
annual_income_range	has_very_low_income	income_source_missing	yearly_income_missing		

tri_morbidity	returned_from_nousing	first_homeless_episode	last_known_nousing_category
last_known_housing_missing	first_intervention_type	days_since_last_activity	

Orange = new Phase B features. All features are extracted from the first snapshot per client — no temporal leakage.

06 Limitations and Honest Caveats

- **Precision remains below target.** At threshold 0.20, precision is 0.377 — below the 0.50 target. Roughly 6 in 10 clients flagged as high-risk did not become chronic in the test set. The model is designed for recall (catching true positives), but caseworkers should be aware of the false alarm rate.
- **Training/test distribution shift.** The training set has a 39.3% chronic rate vs 17.9% in the test set. This reflects real temporal change in the population — newer clients may have shorter follow-up or differ in profile. Performance should be monitored over time.
- **Intervention outcomes are descriptive only.** We cannot establish causal effectiveness of any intervention from observational CAM data. The Goods and Services signal is consistent and noteworthy, but it requires a controlled study to validate.
- **Geographic features excluded.** Geographic region has 47–69% missing data and was excluded from both phases. If location correlates with access to services, this is an uncontrolled confound.
- **External validity.** The model was trained on Waterloo Region data. Performance may differ in other systems with different intake processes or population profiles.

⚠ FAIRNESS ALERT — YOUTH GROUP

The model performs significantly worse for clients under 25 years old. At threshold 0.32, the recall gap between age groups is **0.401** — well above the 0.10 alert threshold.

AGE GROUP	N (TEST SET)	RECALL	FALSE NEGATIVE RATE	STATUS
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Youth (<25)	466	0.551	44.9%	ALERT
Adult (25–55)	1,511	0.877	12.3%	PASS
Senior (>55)	372	0.952	4.8%	PASS

Of every 10 young clients who will actually become chronic, the model only flags 5 — the other 5 leave intake without being identified as high risk. This gap needs to be investigated before relying on the model for decisions involving youth clients. A possible cause is that young clients have different patterns of housing instability and intervention that the current features do not capture well.

07 Recommended Next Steps

PRIORITY	TASK	REFERENCE	PURPOSE
HIGH	Present findings to PECH	06_intervention_outcomes.ipynb	Share the intervention outcomes analysis with PECH in non-technical language — highlight the Goods and Services signal and the selection bias caveat
HIGH	Move app to a shared server	app.py	Currently runs only on a local machine. Moving it to a shared server allows PECH and caseworkers to access it from any computer without technical setup
HIGH	Investigate Youth recall gap	03_fairness_audit.ipynb	The model misses 44.9% of truly chronic youth clients (<25). Investigate whether youth-

MEDIUM	Fairness re-audit with Phase B features	03_fairness_audit.ipynb	Confirm that the new housing and intervention features do not widen recall gaps for Indigenous clients or other groups
MEDIUM	Monitor model performance over time	–	As new data accumulates past the Feb 2024 test window, track ROC-AUC and recall to detect drift from the training distribution

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Model file: phase_b_model1_20260712.pkl · Notebook: 06_intervention_outcomes.ipynb